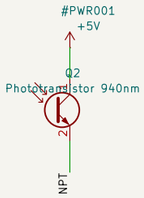
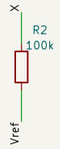
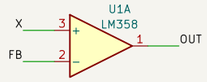
Optical Heartbeat Detector — simple 1-op-amp front end (v2)
PT -> R10k -> C10u -> LM358 gain (VR) -> RC LPF -> A0
Parts: from lab loan box only (LM358, BC548, BC548, phototransistor, IR LED, R/C/VR)



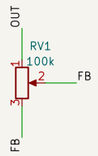






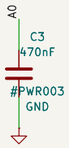


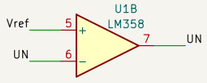








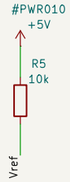


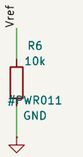


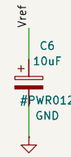


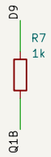


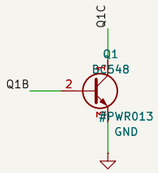


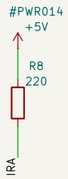




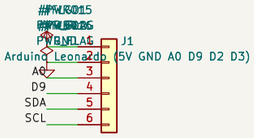


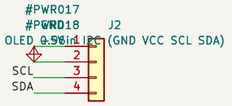












Mini-project 1/2026 — Arduino + LM358 + BC548

Sheet: /
File: heartbeat.kicad_sch

Title: Optical Heartbeat Detector

Size: A3	Date: 2026-10-17	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 1/1